

SCIENTIFIC REPORT ELEVENTH INDIAN EXPEDITION TO ANTARCTICA

TECHNICAL PUBLICATION NO. 9



सत्यमेव जयते

**DEPARTMENT OF OCEAN DEVELOPMENT
CGO COMPLEX, LODI ROAD
NEW DELHI - 110003
INDIA
1995**

Prepared by : Antarctica Division
Geological Survey of India
NH-5 P, N.I.T.
Faridabad-121 001
and
Department of Ocean Development
New Delhi-110003

Printed at : Publications & Information Directorate, CSIR, Dr. K.S. Krishnan Marg,
New Delhi-110 012

FOREWORD

India's scientific foray in Antarctica began in 1981 and the sustained efforts made by our scientists since then have significantly contributed to the knowledge and understanding of this obscure continent which plays a vital role on modulating and sustaining life on earth. Towards this, the members of Eleventh Indian Expedition Team to Antarctica carried out their tasks with utmost dedication despite a number of odds both expected and unexpected.

Commensurate with the growing scientific interests of the nations actively involved in Antarctic exploration, India's field of activity spread to deeper regions of the icy continent during the austral summer of 1991-1992. The geological investigations were extended to Payer-Weyprecht mountains, a relatively unexplored terrain lying 300 kms inland from Antarctic coast and about six thousand feet above sea level. Besides the on-going studies, several experiments on data transfer using HF radio communication between Antarctica and India were conducted successfully. A systematic cataloguing of various avian life forms were undertaken. In addition, a new experiment on lichen growth on various Antarctic substrata was initiated.

In this report, a collection of selected technical papers contributed by the members of the Eleventh Expedition Team and the Ninth Winter Over Team is presented. I appreciate the editorial efforts put forth by Dr. S. Mukerji, the Leader of the Expedition and his colleagues at the Geological Survey of India in bringing out this volume and hope the contents will be extremely useful for furtherance of Polar Science in India.

October, 1995

DR. A.E. MUTHUNAYAGAM

Secretary

Department of Ocean Development
New Delhi-110003

PREFACE

With a decade of experience behind her, India sent a 98 member strong Eleventh Expedition team to Antarctica in the austral summer of 1991-92. Notwithstanding nearly a century of the presence of mankind in that part of our planet, which mostly remains obscure in our drawing room globe; and despite modernization in our approach to have a dialogue with that amazing terrain, Antarctica continues to be as challenging and majestic as ever before. Today, added to the spirit of adventure which glorified the heroic age of Antarctic expedition, is an unquenching thirst for knowledge. Scientific desires and political ambitions of major nations of the world have given a new dimension to exploration activities in Antarctica. For these reasons, every scientific expedition to Antarctica calls for fool-proof planning and execution of various tasks. The foundation laid by previous expeditions was well utilized by the eleventh expedition team, enabling it to brave the natural hazards of Antarctica and emerge successfully with flying colours. The continuation of the scientific work by present expedition which included field-work in more interior areas was an effort largely supported by the pillars of this foundation. The success of such a multi-disciplinary expedition is dependent on support and guidance received from many organisations and individuals to whom I and my team members owe gratitude.

Our effort in Antarctica made amidst all the ingredients that define the spirit of an 'expedition', was inspired by whole hearted support and blessings of Honourable Prime Minister Shri P.V.Narasimha Rao.

Our thanks are due to Prof.V.K.Gaur, Secretary (Retd.), Department of Ocean Development (DOD) and members of the team selection committee for reposing faith in us and giving us an opportunity to work in Antarctica. Various national level scientific organisations viz. Geological Survey of India (GSI), India Meteorological Department (IMD), Indian Institute of Geomagnetism (IIG), National Physical Laboratory (NPL), Zoological Survey of India (ZSI), National Botanical Research Institute (NBRI), Defence Research Development Organisation (DRDO), Banaras Hindu University (BHU), Calcutta University, National Institute of Oceanography (NIO), Survey of India (SOI) and All India Institute of Medical Sciences (AIIMS) participated in the expedition. I thank the heads of all these organisations for maintaining the continuity of scientific work in Antarctica and taking keen interest in Antarctic science. Logistic

support was provided by the three wings of the defence services. My grateful thanks to the Chiefs of Indian Army, Indian Navy and Indian Air Force for their unstinted support during the expedition. I am thankful to the Commandant, Indo-Tibetan Border Police (ITBP), Auli for organising an enjoyable and fruitful pre-expedition training for the team members. Medical Superintendents of Ram Manohar Lohia Hospital and All India Institute of Medical Sciences extended their kind cooperation by completing the medical examination of team members within a short time.

The Ambassador of Columbia in India was among others to see us off at Goa on 27.11.91. His keen interest in the programme of our expedition is appreciated.

On our way to Antarctica we were accorded a very friendly and warm reception at Port Louis by H.E.Shri Rana, India's High Commissioner in Mauritius and by the Chief of Mauritian Police. I am grateful to them for the courtesy shown.

I acknowledge the warmth with which the members of VIII Winter Team led by Dr.A.K.Hanjura received us on our reaching Antarctica.

Mr.George Bush, the then President of U.S.A. and Leader and members of Australian expedition team at Davis station, East Antarctica were among many others who sent us greetings on the occasion of Mid Winter Day,1992.

My sincere thanks to Dr.P.Rama Rao, Secretary, DOD (Retd.) and Dr.A.E.Muthunayagam, Secretary,DOD for their guidance and keen enthusiasm that paved the way to bring out this publication in present form. Shri J.V.R.Prasada Rao, Jt.Secretary and Shri H.P.Rajan, Director (Antarctica) in DOD, New Delhi were a constant source of encouragement for us, all through the expedition, more so, during difficult phases of polar wintering. I extend my sincere gratitude to them.

I acknowledge with thanks the special interest taken in our expedition by Sarvashri C.P.Vohra and S.N.Chaturvedi, former Director Generals of GSI. Grateful thanks are due to Shri D.B.Dimri, Director General of GSI for his keen interest in the outcome of this expedition and its final report.

The Russian stations in Antarctica- Novolazarevskaya and Molodezhnaya; Japanese station Syowa and German stations Georg Forster and G.von Neumayer were in regular touch with us exchanging messages and scientific information. My thanks to the leaders at these stations for keeping up the highest tradition of cordiality in Antarctica.

Smt.Prafulla Prakash of Bombay, an amateur radio (HAM) enthusiast, voluntarily took up the arduous task of remaining in close touch with us at Maitri station through the ether and sending welfare reports to our family members

regularly. Her missionary zeal has contributed handsomely in the success of our expedition.

Shri V.K.Raina, Deputy Director General (Retd.), Shri M.K.Kaul, Director, Antarctica Division and Shri Rasik Ravindra, Senior Geologist of GSI have remained everlasting source of inspiration for me, before, during and after the expedition. I express my personal gratitude to them. Suggestions offered by Shri Ravindra have helped in improving the get-up of this publication.

I record my thanks to H.E.Shri Shyam Saran, Indian High Commissioner in Mauritius, Shri Bidhan Kumar, Counsellor, Indian High Commission and officers and men of Indian naval ship 'Amar' for according us a grand reception at Port Louis on our way back home in March, 1993.

Thanks are due to Chairman, Mormugao Port Trust, Goa; Shri T.V.P.Bhaskara Rao, PSO, Antarctic Study Centre, Goa; Dr.A.Mitra, SSO, DOD, New Delhi and M/s Hiralal & Co., Vasco, Goa for reception and smooth logistic arrangements on our return to India. I also acknowledge the excellent support extended by staff members of OD-I Section, DOD, New Delhi and Antarctic Study Centre, Goa at all times.

Shri Prabh Das, IAS, presently Director (Antarctica), DOD has also contributed in the making of this report through fruitful advice, enthusiasm and showing personal interest in the publication.

I am indebted to Shri M.J.Beg and other colleagues in Antarctica Division, GSI for their generous help and cooperation in the preparatory stages of this report.

Finally, my thanks are due to Captain A.Lennart Hedlund, Master of the ship MV Thuleland and his crew for their assistance, efficiency and cooperation during the expedition.

No amount of effort and planning would have mattered had I not been provided with a team composed of highly motivated and devoted group of scientists and logistic personnel. The ultimate credit for the success of the expedition goes to them.

S. MUKERJI

Antarctica Division
Geological Survey of India
NH - 5P, N.I.T.
Faridabad - 121 001

CONTENTS

	<i>Page</i>
Foreword	v
Preface	vii
<i>S. Mukerji</i>	
The Team	xv
Prologue to the Aims and Achievements of Eleventh Indian Expedition to Antarctica	1
<i>S. Mukerji</i>	

ATMOSPHERIC SCIENCES

Tropospheric Water Vapour Measurement By Microwave Radiometer and to Configure Millimeter Wave Communication System for Antarctica	21
<i>S.K. Datta and A.K.Sen</i>	
Solar Irradiance Measurement and Estimation of Aerosol and Ozone in Antarctica During Austral Summer of 1991- 92	31
<i>S.D. Sharma</i>	
Weather Forecasting in Frigid Desert of Antarctica	43
<i>S.C. Joshi</i>	
Report on the Meteorological Programme Carried Out During Eleventh Indian Expedition to Antarctica	53
<i>A.L. Koppar</i>	
Design and Development of A PC Controlled Acoustic Sounding System for Antarctica	79
<i>N.C. Deb, Jaya Naithani and Mira Kapoor</i>	
Changes in the Auroral Electrojet Currents Inferred From Geomagnetic Field Variations At Maitri and Northern Conjugate Stations	87
<i>Rajesh Kalra, Ajay Dhar, K. Unnikrishnan, K.Jeeva, D.M.Daga and Girija Rajaram</i>	

CONTENTS	<i>Page</i>
HF Communication Between India-Antarctica During 1992 and its Dependence on a) Geomagnetic Activity, b) Season <i>K. Jeeva, A. Hanchinal, Ajay Dhar, D.M. Daga and Girija Rajaram</i>	103
Radio Noise Measurement in HF/VHF Band at Antarctica <i>S.K. Datta</i>	115
 EARTH SCIENCES 	
Geology of the Payer-Weyprecht Mountains, Central Dronning Maud Land, East Antarctica <i>B.R. Bejarniya, V. Ravikant, S. Mukerji, S.C. Nautiyal, L.K. Oberoi and A.S. Gill</i>	121
A Report on the Operations Carried Out by Survey of India Team During Eleventh Indian Expedition to Antarctica <i>P.N. Koul</i>	145
A Note on the Glaciological Studies Carried Out During Eleventh Indian Expedition to Antarctica <i>S. Mukerji, V. Ravikant, B.R. Bejarniya, L.K. Oberoi and S.C. Nautiyal</i>	153
 BIO-SCIENCES 	
On the Avian Forms Encountered During the Eleventh Indian Scientific Expedition to Antarctica <i>Srikumar Chattopadhyay</i>	163
Phytoplankton Ice-edge Blooms in the Marginal Ice Zone at Princess Astrid Coast in Antarctica <i>X.N. Verlencar, V.K. Dhargalkar, S.C. Goswami and N.P. Mhamal</i>	199
Water Uptake and Loss by Antarctic Cyanobacterium <i>Nostoc commune</i> <i>Rajan K. Gupta and A.K. Kashyap</i>	221
Lichen Flora in and Around Maitri Region, Schirmacher Oasis, East Antarctica <i>D.K. Upreti and G. Pant</i>	229

CONTENTS

	<i>Page</i>
Studies on the Use of Cyanobacteria As Biofertilizer For Vegetable Cultivation in Hydroponic System, in Schirmacher Oasis Region, East Antarctica	243
<i>Mohammad Arif, Rajan K. Gupta and M.C. Joshi</i>	
Effect of Cyanobacteria on Various Vegetables Grown in Antarctica	247
<i>Mohammad Arif, Rajan K. Gupta, V.R. Biswas and M.C. Joshi</i>	
Comparative Performance of Thirty Two Crops Grown in Antarctic Soil, Antarctic Moss and Peat Moss Under Glass House Conditions in Antarctica	267
<i>Mohammad Arif and M.C. Joshi</i>	
Occurrence of Invertebrate Fauna in Schirmacher Oasis, Antarctica	301
<i>Mohammad Arif</i>	

MEDICAL SCIENCES

Comparison of Cardiovascular Responses to Cold Stress of Summer and Winterover Members in Antarctica	313
<i>Bimal Chhajer, U. Sachdeva and M. Naidu</i>	
Sleep Studies of Summer Members in Antarctica : Cardiovascular Responses	321
<i>Bimal Chhajer, U. Sachdeva and M. Naidu</i>	
A Study of State Anxiety During Different Periods of Eleventh Antarctic Expedition	329
<i>Debdulal Dutta Roy and Narayan Chandra Deb</i>	

DATA COMMUNICATION

HF Voice and Data Communication Between DEAL, Dehradun and Maitri, Antarctica	337
<i>Kali Ram</i>	
A Note on the Experimental HF Data Link Set Up Between India and Antarctica	341
<i>K.Y. Srikanth</i>	

THE TEAM

Summer Members	Winter Members	Organisation
	Dr.S.Mukerji Leader and Station Commander	Geological Survey of India (GSI)
Dr.B.R.Bejarniya	Shri S.C.Nautiyal	do
Shri V.Ravikant	Shri L.K.Oberoi	do
Shri Amarjeet Singh		do
Maj P.N.Koul		Survey of India (SOI)
Shri B.K.Gairola		do
Shri A.Ravichandran		do
Shri Rajesh Kalra	Shri K.Jeeva	Indian Institute of Geomagnetism (IIG)
Dr.S.D.Sharma		National Physical Laboratory (NPL)
Shri N.C.Deb		Indian Statistical Institute (ISI) / NPL
Shri S.K.Dutta		Institute of Radiophysics and Electronics, Calcutta University
	Dr.A.L.Koppar	India Meteorological Department (IMD)
	Shri S.P.Sharma	do
Dr. V.K.Dhargalkar		National Institute of Oceanography (NIO)
Dr.X.N.Verlencar		do
Dr.D.K.Upreti		National Botanical Research Institute (NBRI)
Dr.Rajan K.Gupta		Banaras Hindu University (BHU)
Dr.S.Chattopadhyay		Zoological Survey of India (ZSI)

Contd.

Summer Members	Winter Members	Organisation
Dr.Bimal Chhajer		All India Institute of Medical Sciences (AIIMS)
Dr.Mohammad Arif		Defence Agricultural Research Laboratory (DARL)
Shri Chandan Bose	Shri K.Y.Srikanth	Research and Development Establishment (R&DE)
Shri D.L.Sapra		do
Shri N.P.Deshmukh		do
Shri V.R.Gujjar		do
Shri Kali Ram		Defence Electronics Application Centre (DEAL)
Shri Uday Shankar		Films Division
Dr.Govind Pant Raju		Journalist
Shri P.S.Singh		Department of Ocean Development (DOD)
Dr.Alberto Dominguez		Colombian Scientist
Capt.A.K.Jindal	Lt Col V.Mishra	Indian Army
Hav P.I.Thommachan	Maj Shiv Sharma	do
Hav P.S.Mahadik	Capt.Suresh Kumar	do
Hav Balwinder Singh	Nb/Sub G.Mani	do
Hav Fateh Bahadur	Nb/Sub Hakam Singh	do
Hav Kapal Singh	Nb/Sub D.S.Subba	do
Hav Yashpal Singh	Hav A.R.Garla	do
Hav Shiv Dayal Singh	Hav G.Pillai	do
HMT R.S.Manhas	Hav M.Baby	do
HMT Yoginder Singh	HMT L.S.Mehta	do
L/Hav M.Nair	HMT Liakat Ali	do
L/Nk K.Prakash	HMT M.T.Patil	do
L/Nk Sree Kumar	HMT Hazara Singh	do
	HMT K.Mukhopadhyay	do
Cdr. A.K. Khanna	Lt Cdr M.S.Prakash	Indian Navy
Lt. Cdr S. Dasika	G.Subba Rao CPOR(TEL)	do
Lt. S.S. Parmar	Rajesh Kumar POELR	do
Lt. P. Rajkumar	D.M.Pillai POCK (O)	do
Lt. S.S. Bal		do

Contd.

Summer Members	Winter Members	Organisation
Lt.A.K.Chowdhary		do
G.R.C.Bhosale MCPOAF II		do
T.D.Sekaran CHAA		do
P.R.Bhosale Mech (AL) 3		do
D.L.Naik EAAR 4		do
A.Shahjee POELR		do
B.Dey LCK (O)		do
P.Yesudas LCK (O)		do
M.Singh CK (O) I		do
Wg Cdr K.P.Singh		Indian Air Force
Wg Cdr S.C.Joshi		do
Sqn Ldr G.S.Cheema		do
Sqn Ldr R.Anthony		do
Sqn Ldr A.Gautam		do
Flt Lt M.R.Anand		do
Flt Lt D.K.Lahiri		do
WO R.R.Yadav		do
WO R.C.Sharma		do
JWO R.Singh		do
JWO V.K.Tiwari		do
JWO Dwivedi		do
Sgt V.Singh		do
Sgt S.D.Sharma		do
Cpl Krishnapotti		do
Cpl V.T.S.Nair		do
Cpl Ramesh Babu		do
Cpl Arun Kumar		do
Wg Cdr A.D.D'Souza	Maj Sujit Ghose	Directorate General Armed Forces Medical Service (DGAFMS)